

COOL:TRACE

The operating system for climate-resilient cooling

Trace the evidence. Prioritize cooling action. Prove what changed.

5 / 10 SITES / ASSETS DEMO	60,656 PUBLIC WEATHER ROWS PUBLIC	175.717 t SCREENED POTENTIAL MODEL	≥10% PILOT ENERGY GOAL TARGET
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An interactive prototype demonstrating an evidence-to-action workflow on preloaded deterministic records. No proprietary hardware is required to start; targeted metering may be required when evidence cannot support a decision or M&V.;

Prepared for Net Zero Challenge 2026 · Track 1

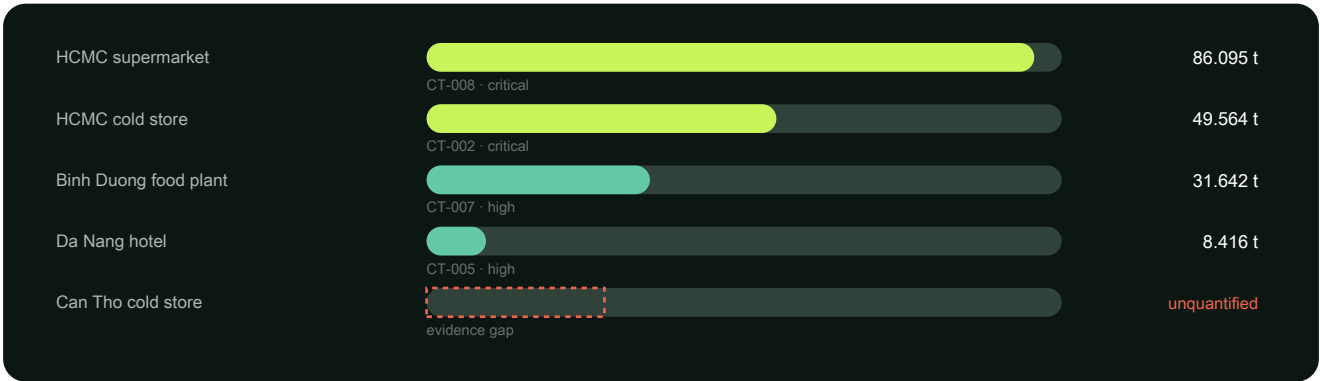
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Current status: interactive prototype in the program's TRL 4–5 category; evidence closest to TRL 4. No AI/ML/LLM, document-content parsing, customer-data persistence, dynamic report generation, customer pilot, or verified field reduction is claimed; only a pre-generated dossier is provided.

Decision brief

Vietnam cannot retrofit every cooling asset at once. The current COOL:TRACE prototype uses preloaded demonstration records to show how facilities could identify evidence-backed inspection priorities, fund a constrained action package, and preserve the evidence needed to prove whether an intervention worked.

94.2 MWh/y ENERGY OPPORTUNITY MODEL	113.621 t DIRECT POTENTIAL MODEL	62.097 t INDIRECT POTENTIAL MODEL	4.03 mo ENERGY PAYBACK MODEL
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The bars are portfolio-screening values from deterministic demonstration records. They are not annual leakage, verified reductions, carbon credits, customer results, or partner claims. CT-010 is intentionally left unquantified because the mass evidence is incomplete.

Decision gate	What the current demo shows	What it refuses to claim
Find	Normalizes preloaded weather, links authored evidence, ranks review	Weather data proves a fault
Fund	Builds a constrained action package from comparable demo invoices	A demo invoice is a market quote
Prove	Replays a fixed M&V case and exposes state and uncertainty	Potential equals verified abatement

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How to Read This Report: Claim Classes

This report makes a strong product case without presenting a demonstration as field performance. Every quantitative claim belongs to one of four classes.

Class	Meaning in this report	Example
PUBLIC	A public source or a deterministic derivative of that source	NASA POWER 41.07°C; official grid emission factor of 0.6592 tCO ₂ /MWh
DEMO	A fixed-seed facility scenario	5 sites, 10 assets, 23 maintenance records
MODEL	An explicit calculation applied to public or demo inputs	CT-002: 2,267.5 kWh modeled avoided
ASSUMPTION	A business input that remains to be tested, such as price, cost, or conversion	Platform subscription price, monthly operating cost, pilot conversion rate

Accordingly, **potential** is a screening value used to compare actions. **verified field reduction** requires facility data and an agreed measurement and verification (M&V) process. The current package contains no verified field reduction.

Report map

- Sections 1–4: Problem, product, and climate/net-zero fit
- Sections 5–9: Data, calculations, Scenario Lab, focus case, and MRV/M&V
- Sections 10–13: KPIs, risks, execution roadmap, and competition fit
- Sections 14–22: Portfolio data, Action Optimizer, economics, product design, architecture, security, evaluation, commercialization, and pitch narrative
- Section 23 and Appendices: Conclusion, key figures, data files, and a one-page executive impact factsheet

1. Executive Summary

Vietnam’s refrigeration and air-conditioning assets create two related climate risks. First, rising outdoor temperatures and hot, humid conditions increase cooling loads, power use, and pressure on operating headroom. Second, the servicing and replenishment of high-global-warming-potential refrigerants such as R-404A can create significant direct greenhouse-gas emissions. In practice, utility data, maintenance invoices, refrigerant top-up records, and weather data often sit in separate systems. Facilities therefore struggle to distinguish normal heat-driven load from asset inefficiency, while repeated refrigerant charging may never become one coherent risk signal.

No proprietary hardware is required to start; targeted metering may be required when existing evidence cannot support a decision or M&V. The current release demonstrates the intended evidence-to-action sequence on preloaded, deterministic records and public climate data.

1. Normalize weather effects with public NASA POWER temperature and humidity data.
2. Resolve asset registers, maintenance records, refrigerant quantities, and power CSVs to individual assets.
3. Screen for repeated charging, unexplained power increases, and shrinking temperature or capacity margins.
4. Use deterministic rules and a transparent optimizer to prioritize the demonstration assets and evidence a technician should review—not to declare a fault or leak.
5. Keep **potential**, **confirmed**, **modeled avoided**, and **verified field reduction** as distinct claim states.
6. Let users test climate stress and operating conditions in an interactive Scenario Lab.

In the focus demonstration case, CT-002, COOL:TRACE links two R-404A top-up records—2.8 kg and 4.1 kg on different dates—into a 6.9 kg historical inventory-imbalance signal. Applying the AR6 GWP of 4,728 specified in Vietnam’s Circular 13/2024/TT-BXD produces 32.623 tCO₂e of direct-emissions screened potential. This is neither a confirmed leak nor a verified reduction. A separate annual electricity-improvement assumption of 8,500 kWh, multiplied by the official 2023 grid emission factor of

0.6592 kgCO₂/kWh, produces 5.603 tCO₂ of indirect-emissions screened potential. Their sum, 38.226 tCO₂e, is a scenario value for prioritization—not a verified total reduction.

For a 15-day common post-repair period, the demonstration compares a signal treated as observed within the scenario, 9,499.6 kWh, with a locked degraded counterfactual of 11,767.1 kWh. The result is 2,267.5 kWh, 19.27%, 1.495 tCO₂, and VND 4,761,647 of modeled avoided energy, emissions, and cost. The scenario carries ±12% relative uncertainty and is not an IPMVP-certified result. A short electricity window cannot verify avoided direct refrigerant emissions, so that claim remains explicitly [not_verified_by_short_energy_window](#).

The product's advantage is not a larger prediction model. It is the combination of:

- Speed: Begin with existing records without proprietary hardware procurement; add targeted metering only where the evidence or M&V boundary requires it.
- Dual value: Address heat resilience and direct plus indirect emissions in one operating workflow.
- Claim integrity: Separate public data, demo facility data, modeled estimates, and field-verified outcomes in both the product and report.

1.1 Current product in numbers

Product asset	Current scope	Status
Public weather observations/reanalysis-derived rows	60,656 rows: 25,568 daily rows and 35,088 hourly rows for HCMC and Bàu Bàng	PUBLIC
National climate scenarios	Historical period plus four future periods under SSP2-4.5 and SSP5-8.5; 9 summary rows total	PUBLIC
Facility evaluation portfolio	5 sites, 4 sectors, 10 refrigeration and air-conditioning assets	DEMO
Maintenance evidence	23 Vietnamese-language records with 23 invoice/work-order references	DEMO
High-resolution operations data	5,472 facility intervals at 15-minute resolution plus 1,368 hours of public weather	DEMO + PUBLIC
Evaluation ground-truth cases	critical 2, high 3, medium 2, low 2, review 1	DEMO
Baseline annual electricity	4,473,000 kWh/year; aggregate nominal capacity 872 kW	DEMO
Screened opportunity	94,200 kWh/year; direct 113.621 tCO ₂ e; indirect 62.097 tCO ₂	DEMO + MODEL

The exact portfolio lineage is 113.6208 direct + 62.09664 indirect = 175.71744 tCO₂e. The displayed components and total are rounded to 113.621, 62.097, and 175.717 only after the underlying calculations; already-rounded operands are not added to produce the total. CT-010 is deliberately excluded because evidence is insufficient. This figure is not a field reduction, annual leak rate, or carbon credit.

1.2 Investment thesis for judges

COOL:TRACE does not solve “the need for another equipment dashboard.” It solves the sequence-of-action problem. It does not treat all 10 demo assets with equal intensity. Three assets—CT-008, CT-002, and CT-007—account for 155.963 tCO₂e, or 88.8%, of total quantifiable screened potential. That concentration tells an operator where a constrained maintenance budget should go first. At the same time, the product correctly de-prioritizes CT-004 and CT-009 as normal weather-driven loads, and it refuses to quantify CT-010 because the evidence is incomplete.

Those three behaviors—concentration, false-positive suppression, and disciplined refusal to quantify—show how an evidence-first decision system can earn operational trust. The current release does not run an AI/ML/LLM model.

2. The Problem

2.1 Heat-driven cooling load can be mistaken for equipment failure

Even healthy equipment uses more electricity when outdoor temperature and humidity rise. A simple month-over-month or before-and-after comparison cannot separate hotter weather, operating hours, throughput, and door-open time from true deterioration. COOL:TRACE converts NASA POWER hourly data to Vietnam local time (UTC+7), joins it to facility electricity data, and builds a baseline that accounts for outdoor temperature, humidity, and operating variables.

2.2 Refrigerant top-ups are rarely connected into one event history

A single refrigerant top-up does not prove a leak. Repeated top-ups, missing recovery quantities, oil traces, pressure anomalies, and follow-up leak-test records on the same asset do, however, create a reason for technician review. The current demonstration links pre-authored structured fields into one auditable evidence chain. AI-assisted OCR and document-field extraction are planned pilot modules, not current capabilities.

2.3 Screened potential and verified impact are often conflated

Credibility collapses when a pre-repair opportunity is presented as a realized reduction. COOL:TRACE enforces four states.

State	Meaning	Permitted language
potential	Opportunity screened from records or an engineering assumption	"inspection priority," "screened potential"
confirmed	A technician record or source document confirms a condition or action	"technician-confirmed," "document-confirmed"; not equivalent to a verified reduction
modeled avoided	A modeled result comparing a locked counterfactual with a post-action signal	"modeled avoided," with uncertainty disclosed
verified field reduction	Confirmed through sufficient field data and an agreed M&V protocol	None in the current demo

2.4 Small and midsize facilities face a high adoption barrier

Solutions that assume dedicated meters and an integrated building-management system create cost and installation delays. The target COOL:TRACE workflow begins with an asset register, maintenance PDFs or images, monthly or 15-minute electricity CSVs, and refrigerant records. No proprietary hardware is required to start, but targeted metering may be required when existing evidence cannot support a decision or M&V. The current release uses preloaded structured records and does not parse those file contents. The core innovation is linking information and designing the follow-up verification, not installing a proprietary hardware fleet.

3. Proposed Service

3.1 Operating principle: Trace the evidence. Prioritize cooling action. Prove what changed.

- Trace the evidence: Preserve source locations, factors, formulas, and missing-data states.
- Prioritize cooling action: Use deterministic screening and budget logic to prepare a technician-review queue, without prescribing physical work.
- Prove what changed: Track performance against a locked post-action baseline and report direct and indirect effects separately.

3.2 Primary users

- Cold-storage and food-processing operators

- Facility teams at supermarkets, hotels, and shopping centers
- HVAC/R service companies and energy service companies (ESCOs)
- Energy, climate, and ESG leaders

3.3 Target core workflow and current release boundary

1. Ingest the asset register and maintenance records.
2. Review candidate refrigerant names, charge and recovery amounts, asset identifiers, dates, and evidence status against their source locations.
3. Join power data with public weather data.
4. Review each asset's risk, efficiency, and evidence-completeness scores.
5. Vary budget and climate-stress assumptions to test the action portfolio.
6. Link technician confirmation and repair outcomes back to source evidence.
7. Track the post-action period, refrigerant balance, and report status.

The current public release demonstrates this intended workflow on preloaded, fixed-seed records. It renders source-linked asset views, runs the deterministic Action Optimizer and Scenario Lab, replays one fixed M&V case, and provides access to a pre-generated competition dossier. Its local file control validates metadata only; it does not parse document contents, run AI/ML/LLM inference, persist customer data, analyze a new facility, or generate a new facility-specific report. AI-assisted OCR and document-field extraction, production connectors, persistence, and dynamic reporting are planned pilot modules; factors, calculations, optimization, claim states, and M&V remain deterministic.

3.4 Target outputs

- Asset-level inspection priorities
- Refrigerant inventory imbalances and missing-evidence list
- Weather- and operations-normalized electricity anomaly screening
- Cooling load, peak demand, and operating-margin views under climate stress
- Cooling Action Pack with direct and indirect emissions separated
- Future facility-specific M&V report with post-action status and uncertainty

The current demo provides the preloaded portfolio and asset screens, deterministic scenario and optimization results, a fixed M&V replay, and access to a pre-generated competition dossier. It does not create these outputs from newly supplied facility records.

4. Climate Change and Net-Zero Fit

4.1 Climate adaptation

A simple OLS screening of annual NASA POWER summaries from 1991–2025 shows a mean-temperature trend of +0.220°C per decade for HCMC and +0.236°C per decade for the Bàu Bàng/Bình Dương proxy location. These are statistical screens for the specified source and period, not standalone proof of causation. The HCMC daily series reports T2M_MAX of 41.07°C on April 30, 2025. NASA POWER is public modeled/reanalysis data based on MERRA-2/GEOS—not a dense network of on-site weather-station measurements.

The World Bank Climate Change Knowledge Portal's Vietnam-wide CMIP6 median shows CDD65 rising from 4,133.28°F-day in the 1995–2014 historical period to 4,868.72°F-day under SSP2-4.5 in 2040–2059, an increase of approximately 17.8%. Under SSP5-8.5 in 2080–2099, the median reaches 6,447.53°F-day, approximately 56.0% above the historical period. These are national scenarios, not forecasts for HCMC or a specific facility.

COOL:TRACE does not use these data to predict a mechanical failure. It uses them to help operators ask: Can the facility maintain its temperature limit under hotter conditions? How close does peak demand move to rated capacity? Which assets should be inspected before a heat event?

4.2 Direct-emissions mitigation

Vietnam’s 2022 HFC consumption survey attributes 69% of total CO₂e consumption to servicing and 954,141 tCO₂e to the refrigeration sector. These are national and sector-level context—not facility leak rates or COOL:TRACE performance. They support the practical case for starting with maintenance records and refrigerant mass balance.

4.3 Indirect-emissions mitigation

Inefficient cooling equipment can increase electricity use and location-based Scope 2 emissions. COOL:TRACE uses Vietnam’s official 2023 grid emission factor of 0.6592 tCO₂/MWh as the default screening factor. This is a national-average, location-based CO₂ factor; it is not a marginal factor, supplier-specific factor, or basis for issuing carbon credits.

4.4 Alignment with Vietnam’s national pathway

- Vietnam’s 2022 NDC targets an unconditional 15.8% reduction from business as usual by 2030 and a 43.5% conditional reduction with international support.
- The Kigali schedule reported in World Bank documentation reduces HFC consumption from a 13.991 MtCO₂e baseline to 2.798 MtCO₂e after 2045, an 80% reduction.
- Vietnam environmental-agency policy context describes a pathway associated with controlled-substance management that reduces 11.2 million tCO₂e by 2045.

These figures are national policy context. None is attributable to COOL:TRACE. The product does not claim achievement of national goals; it improves the evidence and traceability behind facility-level action.

5. Data Design and Provenance Boundaries

The data bundle intentionally separates two provenance zones.

5.1 Public data and deterministic public-data derivatives

Data set	Scope	Role in COOL:TRACE	What it cannot prove
NASA POWER daily	HCMC and Bàu Bàng, 1991–2025	Long-run heat and CDD context	On-site facility temperature
NASA POWER hourly	Two locations, 2024–2025, UTC	Weather covariates for the power baseline	Mechanical failure
World Bank CCKP CMIP6	Vietnam, 1995–2099, SSP2-4.5 and SSP5-8.5	Future cooling-burden scenarios	A specific facility’s future utility bill
Vietnam 2023 grid emission factor	0.6592 tCO ₂ /MWh	Location-based Scope 2 screening	Marginal or supplier-specific emissions
Circular 13/2024 AR6 GWPs	Refrigerant-level GWP100 values	Convert refrigerant mass to direct-emissions potential	That a top-up equals an atmospheric leak
HFC survey, Kigali schedule, and NDC	National and sector level	Policy and market context	Attribution of project performance

All source URLs, retrieval dates, terms of use, spatial and temporal scope, and limitations are recorded in [source_manifest.json](#). Public derivative files can be reproduced with [build_derived.sh](#).

5.2 Demo facility data

The files in [facility_demo/](#) are not customer or partner data. They are deterministic demo scenarios generated with fixed seed 20260821:

- 5 demo sites
- 10 equipment assets
- 23 Vietnamese-language maintenance records
- 5,472 15-minute facility intervals
- Deterioration and repair events with a follow-up M&V result

The 1,368-hour HCMC weather reference comes from public NASA POWER data. Power, throughput, door opening, indoor temperature, equipment status, and repair outcomes are demo signals. In mixed files, [weather_provenance](#) and [facility_signal_provenance](#) preserve that distinction in machine-readable form.

5.3 Derived values and modeled estimates

- Annual climate summaries, OLS trends, and period comparisons: deterministic derivatives of public NASA POWER data
- Heat index and wet-bulb temperature: screening indicators calculated from public weather data
- Facility healthy-load and degraded counterfactual: modeled values using demo coefficients
- Post-action avoided electricity, cost, and Scope 2: comparison of a demo post-action signal and a counterfactual
- Avoided direct refrigerant emissions: none; not verified

6. Analytical Methods and Formulas

6.1 Direct refrigerant-emissions potential

```
direct_potential_tCO2e
= documented_inventory_imbalance_kg × GWP_AR6_100yr ÷ 1,000
```

Facility screening in Vietnam uses AR6 values from Circular 13/2024/TT-BXD. AR4/Kigali values used to describe the policy schedule are not mixed into the same facility result. A top-up is a risk signal. It is not confirmed as an atmospheric release until recovery and transfers within the same boundary have been deducted and the mass balance reviewed.

6.2 Indirect electricity emissions

```
indirect_tCO2
= weather_normalized_kWh_change × 0.6592 kgCO2/kWh ÷ 1,000
```

The output is labeled [location-based Scope 2 screening](#). A real inventory must confirm the latest official factor for the applicable reporting year.

6.3 Demo healthy-load model

```
healthy_power_kW
= 14.8
+ 1.05 × max(Tout - 24, 0)
+ 0.055 × max(RH - 70, 0)
+ 0.0053 × throughput_kg_h
+ 1.10 × door_open_minutes
```

The degraded counterfactual during the common post-action period is [healthy_power](#) × 1.22. The multiplier was not learned from real facility data. It is a demo coefficient that must be refit to historical healthy-operation data and locked before a real deployment.

6.4 Post-action avoided electricity

```
modeled_avoided_kWh
= Σ max(degraded_counterfactual_kW - observed_post_kW, 0) × 0.25h
```

The comparison window runs from May 5, 2025, 14:00 ICT through May 20, 2025, 06:45 ICT. Post-action [observed](#) data are treated as measured signals only within the deterministic demo scenario.

6.5 Cooling stress

```
cooling_stress_degree_hours
= max(outdoor_temperature_c - 24, 0) × interval_hours
```

The lab adds +1.5°C, +2.0°C, or +3.0°C uniformly to the 2025 HCMC public-weather window and reruns the same load equation. This is a sensitivity test—“What if the same operating period were warmer?”—not a downscaled CMIP6 forecast for HCMC and not a probability statement.

6.6 Avoiding double counting

Direct-emissions potential and modeled avoided electricity remain separate. When shown together, the state must be explicit, such as [portfolio screening potential](#); the total cannot appear as a verified reduction. In particular, historical refrigerant top-ups do not automatically become “avoided leaks” after a repair.

7. Scenario Lab Design

Scenario Lab is not a presentation-only dashboard. It is an experimental interface for testing assumptions and sensitivity.

7.1 User-adjustable inputs

- Climate reference: observed/reanalysis-based present conditions or national CCKP scenario context
- Short-run temperature stress: +0.0°C to +3.0°C sensitivity
- Throughput and operating conditions: uploaded values in deployment; scenario values in the demo
- Inspection budget: compare selected assets and screened potential under a spending limit
- Follow-up duration: compare evidence strength for 15-, 30-, or 90-day windows

7.2 Temperature-sensitivity results directly supported by the data

Condition	Modeled energy in common window	Change vs. observed-weather condition	Scope 2	Peak demand	Time at or above 90% of nameplate	Temperature-limit risk time
2025 public weather	18,453.7 kWh	0.00%	12.165 tCO ₂	40.71 kW	0.00 h	0.00 h
Uniform +1.5°C	19,518.9 kWh	+5.77%	12.867 tCO ₂	42.29 kW	0.00 h	0.00 h
Uniform +2.0°C	19,876.9 kWh	+7.71%	13.103 tCO ₂	42.81 kW	0.00 h	0.00 h
Uniform +3.0°C	20,593.0 kWh	+11.59%	13.575 tCO ₂	43.86 kW	2.25 h	0.00 h

These are sensitivity outputs from the demo load equation. They do not prove actual capacity adequacy or failure probability. The first appearance of 2.25 hours above the 90%-of-nameplate alert threshold under +3.0°C does, however, help frame action in terms of operating headroom and service continuity, not only utility savings.

7.3 Recommended output schema

- Adaptation: Expected peak, time near rated capacity, and temperature-limit risk hours
- Mitigation: Direct screened potential, modeled avoided electricity, and location-based Scope 2
- Evidence: Source document ID, calculation factors, model status, and missing data
- Action: Inspection target, technician checks, follow-up evidence, and reassessment date

8. CT-002 Focus Case

8.1 Asset and evidence

Item	Detail	Provenance/status
Asset	CT-HCM-CS-02, compressor bank B at the HCMC demo cold-storage site	Demo facility data
Refrigerant	R-404A; nominal charge 38 kg	Demo asset + official GWP
Maintenance MR-003	February 15, 2025; 2.8 kg top-up; oil trace around service valve; leak inspection recommended	Demo record, follow-up
Maintenance MR-004	April 22, 2025; 4.1 kg top-up; source of loss not identified at the time	Demo record, follow-up
Maintenance MR-005	May 5, 2025; detector inspection, service-valve gasket action, and recovered mass measurement	Demo record, closed
Weather	HCMC NASA POWER hourly values interpolated to 15 minutes	Public-data derivative

8.2 Direct-emissions screened potential

2.8 kg + 4.1 kg = 6.9 kg inventory-imbalance signal
 6.9 kg × 4,728 kgCO₂e/kg + 1,000 = 32.623 tCO₂e

Interpretation: 32.623 tCO₂e is direct-emissions screened potential calculated from historical maintenance records. It does not establish that both top-ups were atmospheric leaks, and it is not a post-repair reduction. The compliant product language is: “Repeated top-ups and efficiency degradation; technician inspection required.”

8.3 Indirect-emissions screened potential

8,500 kWh/year × 0.6592 kgCO₂/kWh ÷ 1,000
 = 5.603 tCO₂/year

The 8,500 kWh value is a demo annual-improvement assumption—not measured savings. Direct 32.623 plus indirect 5.603 equals 38.226 tCO₂e of total screened potential for prioritization.

8.4 Post-action M&V replay

Metric	Result	Status
Post-action demo signal	9,499.6 kWh	Demo facility data
Locked degraded counterfactual	11,767.1 kWh	Modeled estimate
Modeled avoided electricity	2,267.5 kWh	modeled avoided
Modeled reduction rate	19.27%	modeled avoided
Modeled location-based Scope 2 avoided	1.495 tCO ₂	modeled avoided
Modeled avoided cost	VND 4,761,647	Demo assumption of VND 2,100/kWh
Relative uncertainty	±12%	Scenario assumption
Method status	Demonstration M&V	Not IPMVP-certified
Avoided direct refrigerant emissions	Not verified	A short electricity window cannot verify this claim

8.5 Next evidence gates

1. Mass balance reconciling recovery, recharge, and inventory within one boundary
2. Technician-signed record documenting the serviced component and test method
3. Evidence of no repeated top-up during the agreed follow-up period
4. Records of normal operation, throughput, door opening, and setpoint changes
5. Review of the pre-specified baseline and uncertainty

Direct refrigerant reductions cannot advance to **verified** before those gates are satisfied.

9. Measurement, Reporting, and Verification (MRV) / Measurement and Verification (M&V) Design

9.1 Before action

- Document the asset and meter boundary and refrigerant inventory boundary.
- Agree on the baseline period, exclusions, and variables such as temperature, humidity, throughput, and operating hours.
- Lock the GWP assessment basis and grid-factor year.
- Lock all pre-action calculations as **potential**.

9.2 During action

- Have the technician record the finding, leak location or non-leak cause, quantities recovered and charged, and corrective action.
- Distinguish leak repair, cleaning, sensor calibration, and other action types.
- Require every future AI-assisted candidate field to remain traceable to an exact source-document location and pass human review.

9.3 After action

- Collect data under the same or comparable operating conditions.
- Calculate the counterfactual with the model locked before action.
- Report the follow-up duration, completeness, uncertainty, and exclusions with the result.
- Verify direct refrigerant effects through a separate mass-balance and recurrence-prevention evidence path—not through electricity M&V.

9.4 Proposed status by verification stage

Stage	Recommended product status	Meaning
Before action	Potential	Used to prioritize action
Technician record obtained	Confirmed finding	Condition or action confirmed; reduction remains unverified
Short post-action window	Modeled avoided / provisional	Material seasonal and operating-variation risk
Sufficient follow-up + data-quality pass	Verification review eligible	Ready for independent review or agreed M&V
Approved method and review complete	Verified field reduction	Outside the current demo scope

10. Key Performance Indicators

10.1 Planned pilot product and data KPIs

- User-confirmed accuracy of future AI-assisted refrigerant, mass, asset, and date candidate fields
- Share of claims traceable to a source document or calculation
- Completeness of provenance tags across public and facility data
- Success rate for UTC-to-UTC+7 conversion and power-data alignment
- Accuracy of cases in which the system refuses to quantify because evidence is missing
- Cases in which an unconfirmed leak is presented as confirmed: target 0

10.2 Planned pilot operating KPIs

- Time from receipt of supported records to a human-reviewed prioritized inspection list
- Share of assets connected to evidence
- Confirmed high-priority findings per technician visit
- Share of completed actions with follow-up data
- Time from alert to inspection completion

10.3 Climate-adaptation KPIs

- Hours near or above 90% of nameplate capacity
- Temperature-limit exceedance or risk hours
- Share of preventive inspections completed before heat events
- Peak load and operating headroom by stress scenario

10.4 Mitigation KPIs

- Separate totals for direct refrigerant **potential** and **verified**
- Weather- and operations-normalized modeled avoided electricity
- Modeled location-based Scope 2 avoided
- Completeness of uncertainty, verification status, and factor basis for each result

Demo values are not evidence that business KPIs have been achieved. Field-pilot baselines must be established independently.

11. Risks, Limitations, and Controls

Risk	Description	Control
Weather-data overinterpretation	NASA POWER is not an on-site sensor	Label as public modeled/reanalysis ; separate from site temperature
Climate-scenario overinterpretation	CCKP is a national spatial-average scenario	Prohibit facility forecasts and probability language
Refrigerant leak misdiagnosis	Top-up mass does not equal leak mass	Technician-confirmation gate; review recovered mass and boundary
GWP-basis mixing	AR6 and AR4/Kigali values differ	Show basis on every result; never mix bases within one result
Scope 2 overstatement	National-average factor is not marginal emissions	Label as location-based screening
Overfit or post-hoc baseline changes	A model can be adjusted after viewing the result	Lock baseline in advance; record version and exclusions
Short post-action window	May not capture seasonality or operational variation	Provisional status with uncertainty
Demo mistaken for a pilot	Facility data are synthetic scenarios	Persistent demo provenance in product and report
Unsafe automated repair advice	HVAC/R work requires expert judgment	Current release uses no AI; any future AI assistance is limited to source-linked candidate document fields
Changing regulation or factors	Applicability and official values can change	Confirm current law and factors before reporting

COOL:TRACE does not replace legal advice, refrigerant-safety procedures, carbon-credit verification, or technician judgment.

12. Execution Roadmap

Phase 1 — Complete the demonstration

- Place Scenario Lab inputs, formulas, and sources on the same screen.
- Complete the CT-002 evidence chain and post-action replay.
- Distinguish public, demo, model, and unverified states with text as well as color.
- Support printable reporting and scenario JSON export.

Phase 2 — Validate data connectivity

- Map each facility's asset-register and maintenance-document templates.
- Test monthly or 15-minute electricity data quality.
- Validate time zone, missingness, units, and asset boundary.
- Fit and lock a healthy-load baseline to real facility data.

Phase 3 — Limited field pilot

- Evaluate prioritization accuracy on a small asset set at one facility.
- Use technician decisions as ground truth and record false positives and false negatives.
- Agree on the pre/post-action M&V plan before action.
- Track direct refrigerant and electricity effects separately.

Phase 4 — Portfolio operations

- Multi-asset workflows for facilities and HVAC contractors
- Factor, GWP, and model version control with audit logs
- Climate-stress-based preventive maintenance calendar
- Evidence packages suitable for external review

The current data bundle does not claim an actual pilot, partner commitment, or technician review.

13. Official Fit with Net Zero Challenge 2026

13.0 Official program criteria

Net Zero Challenge 2026 officially opened on July 1, 2026. It requires an MVP, a plan to demonstrate proof of value through a pilot in Vietnam, and an intent to commercialize. The official site presents the evaluation perspectives as Tech Domains, Adoption & Scalability, Vietnam Relevance, and Implementation Plans. The 2026 application window runs from July 1 through August 31, and the Grand Finale is November 12. [Official 2026 site](#) · [2026 launch announcement](#)

The official application includes [Prototype / MVP \(TRL 4–5\)](#) as a selectable maturity stage. It asks for the project overview, Vietnam pilot objectives, metrics, statement of work, timeline, budget, and partner status. COOL:TRACE’s accurate form selection is this combined category; current evidence is closest to TRL 4, and the proposed pilot is intended to validate the system in a relevant field environment. The team has an interactive prototype and fixed evaluation data, but no local pilot, field validation, or revenue. [Official application form](#) · [Official FAQ](#)

COOL:TRACE will apply to Track 1 — Renewables & Carbon Removal. The 2026 launch announcement includes energy efficiency and carbon measurement in Track 1. The official Challenge Statement expressly cites industrial and building emissions, including [cooling systems](#), and seeks technology and business solutions for energy efficiency. [Official Challenge Statement](#)

13.0.1 Response to the official evaluation dimensions

Official dimension	COOL:TRACE response	Submission evidence	Remaining weakness and mitigation
Technology / tech domain	Current source-linked deterministic evidence model, weather-normalized time series, claim engine, Scenario Lab, and optimizer; planned AI-assisted OCR/extraction	Interactive web prototype, public-data bundle, formulas, model card, 10-case regression set	No current content parsing, trained AI model, or blind extraction-accuracy result
Adoption & scalability	No proprietary hardware required to start; targeted metering where evidence requires it; contract unit of 5 sites/100 assets	Five-site, 10-asset data contract; modular architecture	Onboarding time and willingness to pay remain untested
Vietnam relevance	Official grid factor, Circular 13 GWP values, NASA/CCKP climate data, Vietnamese maintenance documents	60,656 weather rows; HFC, Kigali, and NDC context	No local facility sensors, tariff data, or technician review
Implementation / pilot impact	Defined nine-month scope: two anchors first, conditional expansion to a maximum of six after Gate B	Pilot plan, targets, and stop conditions below	No current partner or pilot; partner acquisition is the first gate

13.0.2 Proposed nine-month pilot: two anchors first, conditional expansion to six

This is a planned pilot target to be executed if supported. It is not a secured pilot or partner commitment.

Period	Facility scope	Major work	Decision gate
Month 0–1	Secure 2 anchor candidates	Contracting, data boundaries, safety responsibility, asset map, baseline design	Confirm partners, data access, and technician responsibility
Month 1–3	Two anchor facilities	Join maintenance records, refrigerant records, power, and weather; silent evaluation; technician review; baseline pre-lock	Data completeness $\geq 90\%$; evidence linkage 100%; minimum M&V screen CV(RMSE) $\leq 25\%$ and absolute NMBE $\leq 10\%$
Gate B / Month 4–5	Conditional expansion to up to 4 additional facilities	Review partner, quotation, data, safety, technician, and baseline evidence; expand only after approval	Stricter Gate-B target CV(RMSE) $\leq 20\%$ and absolute NMBE $\leq 5\%$; both tiers reviewer-approved and pre-locked
Month 4–8	Eligible participating facilities	Complete approved action and track post-action power, refrigerant mass, temperature, and operating data	Locked facility-level M&V result with uncertainty or explicit inconclusive state
Month 9	Full portfolio	Prepare independent review; make commercialization and scale decision	Go/no-go; pricing, ROI, and expansion contract

Target facility mix: two cold-storage facilities, two food-processing plants, one supermarket, and one hotel or commercial facility. This is a target mix for cross-sector evaluation—not a completed recruitment result.

Primary outcome target: For eligible facilities that complete an intervention and maintain a stable metering and operating boundary, target a $\geq 10\%$ reduction in weather- and operations-normalized cooling electricity. This is a pilot success criterion, not a promised savings rate.

Guardrails for the 10% target

- Lock the baseline before observing results and record the healthy period, exclusions, and variables.
- Apply $\geq 10\%$ only to the normalized cooling-meter boundary—not to total electricity or a simple before/after comparison.
- If minimum data completeness, holdout error, or operating-range conditions fail, classify the result as **inconclusive**.
- Keep direct refrigerant reductions outside the electricity target and verify them separately through mass balance and evidence of non-recurrence.
- Do not require all participating facilities to reach 10%. Report the distribution, confidence intervals, misses, and failures for eligible facilities.
- Do not declare success if action cost, production, temperature limits, or service continuity deteriorate, even if the energy target is met.

13.0.3 Local partner status and acquisition plan

Current status: GEONOS Infrastructure does not have a signed pilot agreement, technician validation letter, or anonymized real maintenance data from a Vietnamese cold-storage operator, hotel, food plant, or supermarket. The application should state: **current partner: none; partner acquisition in progress**.

Required partner roles

Role	Minimum first responsibility	Acquisition approach
Facility host	Asset, power, and maintenance data; site access; action approval	Approach cold-chain, food, retail, and hotel associations and industrial-park operators
HVAC/R service company	Safety inspection; confirm or reject leak hypothesis; action record	Recruit two firms to avoid dependence on one brand or vendor
Energy/M&V reviewer	Review baseline, meter boundary, and uncertainty	Limited-scope engagement with a local ESCO or energy-audit professional
Program/industry connector	Introduce facility candidates and support pilot administration	Request introductions through the NOC corporate/government partner network

Role	Minimum first responsibility	Acquisition approach
Legal/data advisor, if needed	Review data handling, site responsibility, and applicable regulation	Obtain local expert review before contracting

Acquisition sequence: 20-candidate long list → 10 discovery calls → 6 data-fit assessments → 2 anchor letters of intent → 6 facility statements of work. If partners are not yet secured, the team will not fabricate evidence. It will make a specific request for the introductions needed through the Challenge network.

13.1 Clearly defined climate problem

Cooling creates indirect emissions through electricity use and direct emissions through refrigerants. COOL:TRACE keeps the evidence and calculation rules separate instead of optimizing one side and ignoring the other.

13.2 Software-only implementation

The service begins with existing maintenance records and power CSVs, so it does not wait for large equipment replacement. It augments HVAC/R technicians by improving visit prioritization and evidence completeness rather than replacing their judgment.

13.3 Adaptation and mitigation in one workflow

Scenario Lab goes beyond utility savings. It shows peak demand, rated-capacity headroom, and temperature risk under warmer conditions—connecting climate action to food, pharmaceutical, and indoor-environment service continuity.

13.4 Data transparency

Every material number links to a source document, public source, calculation, and claim state. The refusal to present a synthetic demonstration as actual impact is itself a trust feature for financial, carbon, and ESG decision-making.

13.5 Scalability

The data model can serve cold storage, food processing, supermarkets, hotels, shopping centers, and HVAC service portfolios. Demo performance is not assumed to generalize across sectors; each equipment class and sector requires its own baseline and validation.

13.6 Required demonstration sequence

1. Move the climate-stress control from +0°C to +3°C.
2. Show modeled electricity increasing from 18,453.7 kWh to 20,593.0 kWh and peak demand from 40.71 kW to 43.86 kW.
3. Open CT-002’s 2.8 kg and 4.1 kg top-up records and trace the 32.623 tCO₂e screened potential to its source.
4. Show [technician inspection required](#), never [AI-confirmed leak](#).
5. Show 2,267.5 kWh modeled avoided after action together with ±12% uncertainty.
6. Confirm in the report that avoided direct refrigerant emissions remain unverified.

That sequence makes clear that COOL:TRACE is not a decorative carbon dashboard. It is an evidence-to-action-and-verification service.

14. Full Portfolio Data and Evaluation Cases

CT-002 is a focus case used to explain the evidence chain. A real company product must apply the same data contract and claim rules across a portfolio of sites, sectors, refrigerants, and priorities. The current demonstration uses a fixed evaluation set of five sites and 10 assets for that purpose.

14.1 Five-site portfolio

Site	Sector	Assets	Nominal capacity	Baseline annual electricity	Nominal refrigerant charge	Quantified screened potential
SITE-HCM-CS	HCMC cold storage	3	160 kW	990,000 kWh	147 kg	49.564 tCO ₂ e
SITE-DN-HT	Đà Nẵng hotel	2	190 kW	738,000 kWh	137 kg	8.416 tCO ₂ e
SITE-BD-FP	Bình Dương food processing	2	320 kW	1,605,000 kWh	462 kg	31.642 tCO ₂ e
SITE-HCM-SM	HCMC supermarket	2	160 kW	912,000 kWh	113 kg	86.095 tCO ₂ e
SITE-CT-CS	Cần Thơ cold storage	1	42 kW	228,000 kWh	31 kg	Quantification withheld
Total	4 sectors, 5 sites	10	872 kW	4,473,000 kWh	890 kg	175.717 tCO ₂ e

Of the 890 kg nominal refrigerant charge, 420 kg is R-717, which has a GWP of 0 under the applied factor basis. Refrigerant mass alone therefore cannot determine priority. Direct climate impact depends on refrigerant type, mass balance, and evidence state.

14.2 Exact 10-asset evaluation table

Case	Asset and core finding	Priority	Direct potential	Annual electricity opportunity	Indirect potential	Total potential	Evidence confidence
CT-001	HCMC cold-storage compressor A — condenser fouling	High	0	12,000 kWh	7.910 tCO ₂	7.910 tCO ₂ e	0.91
CT-002	HCMC cold-storage compressor B — repeated charging and efficiency degradation	Critical	32.623 tCO ₂ e	8,500 kWh	5.603 tCO ₂	38.226 tCO ₂ e	0.96
CT-003	HCMC loading-zone evaporator — door opening and gasket	Medium	0	5,200 kWh	3.428 tCO ₂	3.428 tCO ₂ e	0.88
CT-004	Đà Nẵng hotel chiller — normal heat-wave load	Low	0	0	0	0	0.94
CT-005	Hotel VRF — repeated charging, location unconfirmed	High	6.768 tCO ₂ e	2,500 kWh	1.648 tCO ₂	8.416 tCO ₂ e	0.82
CT-006	Bình Dương glycol chiller — sensor calibration	Medium	0	0	0	0	0.93
CT-007	Food-plant NH ₃ system — heat-exchanger scale	High	0	48,000 kWh	31.642 tCO ₂	31.642 tCO ₂ e	0.91
CT-008	HCMC supermarket refrigeration rack — repeated charging and efficiency degradation	Critical	74.230 tCO ₂ e	18,000 kWh	11.866 tCO ₂	86.095 tCO ₂ e	0.97
CT-009	Supermarket rooftop AC — normal extreme-heat load	Low	0	0	0	0	0.95
CT-010	Cần Thơ condensing unit — missing scale photo and cylinder reconciliation	Review	Quantification withheld	0	0	Quantification withheld	0.45
Quantifiable total	6 nonzero opportunities; 3 direct-emissions risks	—	113.621 tCO ₂ e	94,200 kWh	62.097 tCO ₂	175.717 tCO ₂ e	—

The evaluation set is not composed only of anomalies. It includes two normal weather-load cases, one sensor issue, one infiltration/operations issue, and one insufficient-evidence case. These cases test the deterministic ranking, abstention, and claim-language rules—not an AI model. In all 10 cases, may_call_leak_confirmed_without_technician_record is false.

14.3 Data volume and join structure

Layer	Rows/count	Join key	Product role
NASA POWER daily	25,568 rows	Location and local date	Long-run heat, CDD, and 1991–2025 trend
NASA POWER hourly	35,088 rows	Location and UTC time → ICT	Weather-normalized baseline
CCKP CMIP6 summary	9 rows	Scenario and period	Long-run cooling-burden experiment
Facility 15-minute series	5,472 rows	asset_id and ICT timestamp	Power, throughput, door opening, indoor temperature
Focus-window public weather	1,368 rows	UTC timestamp	Weather anchor for 15-minute interpolation
Asset register	10 rows	asset_id, site_id	Refrigerant, nameplate, setpoint, annual electricity
Maintenance records	23 rows	record_id, asset_id, date	Source evidence, mass, action, status
Fixed evaluation ground truth	10 cases	case_id, asset_id	Regression tests and overclaim detection

The web application does not render every raw row simultaneously, but its summaries are calculated from this join structure. Source and derivative files preserve hashes, versions, retrieval dates, and provenance.

15. Action Optimizer — Turning a Constrained Budget into an Action Sequence

15.1 Why optimization is needed

Most facilities cannot investigate every asset at once. COOL:TRACE keeps climate potential, evidence readiness, resilience, economics, and urgency visible, but the current demonstration optimizer does not combine them into an opaque weighted score. Eligibility is determined before optimization; the selected lens then uses one inspectable objective.

1. Climate impact: Direct and indirect screened potential
2. Evidence readiness: Completeness of source, mass, asset, and date
3. Resilience: Time near peak capacity and temperature-limit risk
4. Economics: VND value of the electricity opportunity and proxy action cost
5. Urgency: Critical, High, Medium, Low, or Review

The current release exhaustively evaluates every subset of the five eligible actions—32 candidate packages. It is deterministic and is not an AI/ML/LLM recommendation model.

```
Climate lens:
maximize Σ raw totalPotentialTco2e_i × selected_i

Cash lens:
maximize Σ annualElectricitySavingVnd_i × selected_i

subject to Σ action_cost_proxy_i × selected_i ≤ user_budget
```

Only CT-001, CT-002, CT-003, CT-007, and CT-008 enter the comparable set because each has a positive modeled energy opportunity, a non-null action cost, and a demonstration invoice whose scope matches the modeled action. CT-005 is excluded because its VND 2.3 million record covers a diagnostic or evidence-closure step, not a scope-matched repair. CT-006 and CT-009 have no modeled carbon saving; CT-010 lacks sufficient mass evidence and a future-action cost.

For an equal primary score, the algorithm prefers lower total cost. An exact score-and-cost tie retains the first subset encountered in the fixed input order. There is no hidden secondary objective. Safety, legally required inspections, and technician judgment remain outside the budget calculation.

15.2 Demo cost proxies

The following amounts reuse action-invoice values from the scenario's maintenance records. They are demo proxies, not quotations, market prices, or guarantees.

Asset	Cost proxy	Evidence record	Included scope	Raw screened potential	Annual electricity-opportunity value*
CT-001	VND 4.1M	MR-002 cleaning	Condenser cleaning	7.910 tCO ₂ e	VND 25.20M
CT-002	VND 14.6M	MR-005 leak test & repair	Leak test and service-valve action	38.226 tCO ₂ e	VND 17.85M
CT-003	VND 3.7M	MR-007 door seal	Gasket and door-time adjustment	3.428 tCO ₂ e	VND 10.92M
CT-007	VND 18.4M	MR-016 descaling	Heat-exchanger descaling and pump adjustment	31.642 tCO ₂ e	VND 100.80M
CT-008	VND 23.8M	MR-019 leak test & repair	Evaporator connection repair	86.095 tCO ₂ e	VND 37.80M

CT-005 remains visible in the evidence queue, but not in optimizer packages. Its VND 2.3M record is an inspection cost. It does not support a repair payback or selection of the 8.416 tCO₂e screen as a fundable repair package.

15.3 Deterministic results by budget

Budget cap	Lens	Selected action portfolio	Proxy cost	Raw screened potential	Annual electricity opportunity	Annual electricity-opportunity value*
VND 10M	Climate or cash	CT-001, CT-003	VND 7.8M	11.338 tCO ₂ e	17,200 kWh	VND 36.12M
VND 25M	Climate	CT-008	VND 23.8M	86.095 tCO ₂ e	18,000 kWh	VND 37.80M
VND 25M	Cash	CT-001, CT-007	VND 22.5M	39.552 tCO ₂ e	60,000 kWh	VND 126.00M
VND 50M	Climate	CT-001, CT-002, CT-003, CT-008	VND 46.2M	135.659 tCO ₂ e	43,700 kWh	VND 91.77M
VND 50M	Cash	CT-001, CT-003, CT-007, CT-008	VND 50.0M	129.075 tCO ₂ e	83,200 kWh	VND 174.72M
VND 60M	Climate	CT-002, CT-007, CT-008	VND 56.8M	155.963 tCO ₂ e	74,500 kWh	VND 156.45M
VND 60M	Cash	CT-001, CT-003, CT-007, CT-008	VND 50.0M	129.075 tCO ₂ e	83,200 kWh	VND 174.72M
VND 70M	Climate or cash	All five eligible actions	VND 64.6M	167.301 tCO ₂ e	91,700 kWh	VND 192.57M

* Annual opportunity value uses a demo assumption of VND 2,100/kWh. It is not measured savings.

15.4 State machine and safeguards

```

screened
→ evidence_requested
→ technician_confirmed_or_rejected
→ action_completed
→ M&V_provisional
→ verification_review_eligible
→ verified_field_reduction

```


- A finding cannot jump automatically from **screened** to **action_completed**.
- **insufficient_evidence** is not zero; it is a do-not-generate-a-number state.
- Normal weather-driven load is de-prioritized, while safety and temperature-limit alerts remain active.
- Direct refrigerant and indirect electricity values enter an aggregate view only after their own verification gates.
- The Optimizer does not replace technician safety judgment, legally required inspections, or refrigerant-handling procedures.

16. Portfolio Economics and Business Unit Economics

16.1 Facility economics screen

Annual electricity opportunity comes from demo ground-truth assumptions. Costs are proxies drawn from the scenario action invoices. Simple payback is a prioritization aid, not an investment-grade financial decision.

Asset	Proxy action cost	Annual electricity opportunity	Value at VND 2,100/kWh	Simple payback screen
CT-001	VND 4.1M	12,000 kWh	VND 25.20M	2.0 months
CT-002	VND 14.6M	8,500 kWh	VND 17.85M	9.8 months
CT-003	VND 3.7M	5,200 kWh	VND 10.92M	4.1 months
CT-005	VND 2.3M	2,500 kWh	VND 5.25M	Withheld—inspection cost only
CT-007	VND 18.4M	48,000 kWh	VND 100.80M	2.2 months
CT-008	VND 23.8M	18,000 kWh	VND 37.80M	7.6 months

The five cases with complete action proxies—CT-001, CT-002, CT-003, CT-007, and CT-008—sum to VND 64.6M of proxy cost, 91,700 kWh of annual electricity opportunity, VND 192.57M of annual opportunity value, and approximately 4.0 months of simple payback. This is a demo economics screen, not realized post-repair performance.

For CT-002, the evidence currently supports only the VND 4,761,647 modeled avoided during the 15-day common window, with $\pm 12\%$ uncertainty. The short window is not extrapolated mechanically to 365 days.

16.2 Five layers of customer value

Value layer	Quantified in current demo?	Evidence required in a real pilot
Electricity-cost savings	Partially; modeled/assumed	Meter boundary, locked baseline, tariff, follow-up period
Avoided refrigerant purchase/disposal	No	Purchase price and recovery, reuse, and disposal records
Avoided food loss/downtime	No	Production interruption, quality, and temperature-excursion records
Reduced technician travel and diagnosis time	No	Labor time and first-time-fix rate
Reduced reporting and audit-preparation time	No	Report-production time and missing-evidence rate

Carbon prices and credit revenue are excluded. Methodology eligibility, additionality, verification, and ownership would require separate work.

16.3 GEONOS Infrastructure's initial pricing experiment

The following is an ASSUMPTION for commercial design—not a quotation, contract, or market-validation result.

Item	Assumption	Validation method
Target contract unit	One portfolio with up to 5 sites and 100 assets	Confirm data and organizational boundaries in customer interviews
Initial data connection and baseline setup	VND 25M one-time	Measure actual onboarding labor
Platform and evidence monitoring	VND 10M/month	Willingness-to-pay interviews and paid pilot
Monthly variable cost	VND 3M/month	Assumption: cloud 0.9M, OCR/models 0.8M, support/QA 1.3M
Recurring contribution profit	VND 7M/month	Track actual monthly cost of goods sold
Recurring gross margin	70%	Re-estimate after five paying customers
Direct implementation cost	VND 15M	Meter data-mapping and training time
Year-one contract value	VND 145M	$25M + 12 \times 10M$
Year-one contribution profit	VND 94M	$10M \text{ implementation} + 84M \text{ recurring}$
Customer-acquisition cost assumption	VND 35M	Temporary value before measuring channel and selling costs
CAC payback on recurring contribution	5 months	$35M \div 7M/\text{month}$

These unit economics do not claim a validated business. They convert pilot learning into measurable questions. Price, support intensity, model cost, and retention will be re-estimated after the first five paying customers.

16.4 Revenue-model priorities

1. Portfolio Evidence Scan: One-time data normalization and opportunity map
2. Continuous COOL:TRACE: Subscription monitoring, Action Inbox, and reporting
3. M&V Pack: Action-level baseline lock, follow-up tracking, and external-review package
4. Service Partner Console: Multi-customer evidence and work management for HVAC/R companies

The initial product charges for data and workflow—not for selling emissions reductions or guaranteeing performance.

17. Product Design as a Real Company and Service

17.1 Company positioning

GEONOS Infrastructure is a software company that converts climate data into action decisions for physical infrastructure. COOL:TRACE is its first vertical product, connecting climate resilience, energy, and refrigerant evidence in one operating system for cooling facilities.

Brand promise: **Trace the evidence. Prioritize cooling action. Prove what changed.**

Product category: **Evidence-gated cooling intelligence and M&V workflow.**

17.2 Target operating-product modules

The table below describes the target operating product. The current release implements the preloaded Portfolio Command, Scenario Lab, deterministic Action Optimizer, fixed M&V replay, and access to a pre-generated dossier. Evidence Inbox content extraction, persistence, workflow assignment, dynamic reporting, and tenant administration are not implemented.

Module	Primary interface	Job completed by the user
Portfolio Command	Site/asset risk map and data completeness	Decide where to look first
Evidence Inbox	Maintenance documents, refrigerant mass, source links	Review planned AI-assisted candidate fields and request missing evidence
Climate Context	Current heat, CDD, and CMIP6 scenarios	Separate weather load from structural risk
Scenario Lab	Temperature, throughput, budget, and verification controls	Stress-test the action portfolio
Action Optimizer	Selection, constraints, and sensitivity by budget	Allocate maintenance budget
M&V Ledger	Baseline version, counterfactual, and uncertainty	Track post-action results
Report Studio	Executive, technical, and ESG views; JSON/PDF	Produce an auditable decision package
Admin & Factors	Factors, GWP values, access, audit log	Govern calculation standards and permissions

17.3 Buyers and users

Role	Core question	COOL:TRACE decision support	Success metric
COO / plant manager	What could disrupt operations during heat?	Assets prioritized for service continuity	Temperature excursions and downtime
CFO / asset owner	Where should constrained capital go?	Action portfolio by budget	Approval time and recoverable opportunity
Energy / ESG lead	Which numbers are reportable?	Results with state, factor, and uncertainty	Evidence linkage and rework
Facility manager	What should be checked today?	Action Inbox with evidence and checklist	Alert-to-inspection time
HVAC/R technician	What is the hypothesis and source history?	Asset-level maintenance and refrigerant timeline	First-time-fix rate
External reviewer	Can the calculation be reproduced?	Source, version, formula, and exclusion package	Questions and rejected submissions

17.4 Core workflows

The following workflows describe the target operating product unless a step is explicitly identified as part of the current preloaded demonstration.

Buyer: 15-minute portfolio review

1. Review critical/high items and data gaps across five sites.
2. Move the budget from VND 25M to VND 60M and inspect changes in the action portfolio.
3. Review direct refrigerant and indirect electricity potential separately.
4. Check the evidence status and whether technician confirmation is required for each action.
5. Download an approval package and assign an owner and due date.

Facility manager: daily Action Inbox

1. Ingest a new invoice or work record, or import it from an inbox.
2. Compare planned AI-assisted candidate asset, refrigerant, mass, and date fields with the source.
3. Classify the case as normal weather load, efficiency issue, refrigerant risk, or insufficient evidence.
4. Send inspection checks to a technician and capture the site result.

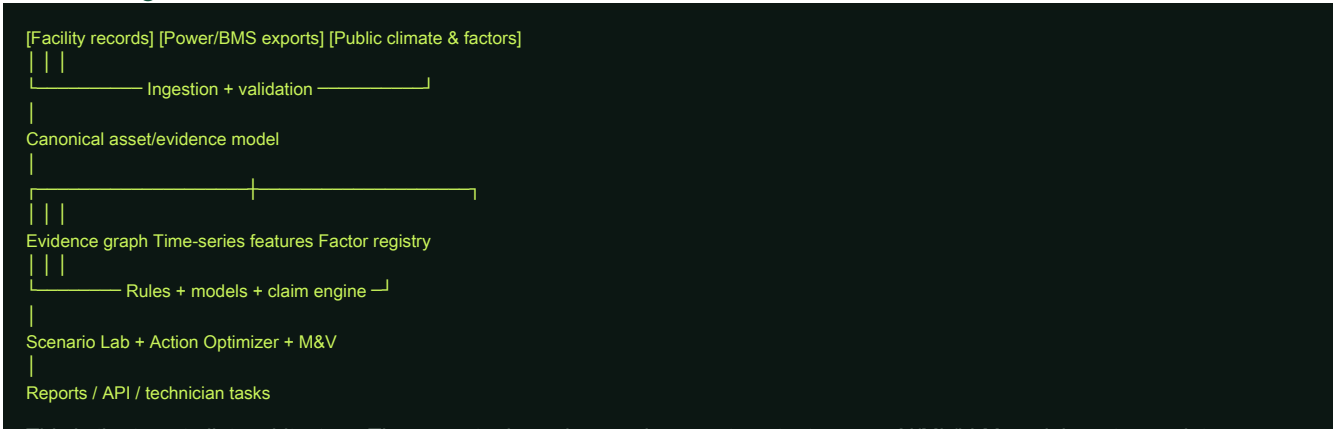
5. Lock the baseline on the action-completion date and start follow-up.

Technician: site visit

- 1. Open the historical evidence chain by asset QR code or `asset_id`.
- 2. Review the hypothesis to test and evidence gaps, not an automated or AI verdict.
- 3. Record method, quantities recovered and charged, component, test result, and photo.
- 4. Sign the status: leak confirmed/rejected, efficiency action, or safety action.
- 5. Only then may the system advance the claim state.

18. Product Architecture and Data Pipeline

18.1 Target architecture



This is the target pilot architecture. The current release has no document-content parser, AI/ML/LLM model, customer-data persistence, production connector, tenant workflow, or dynamic report generator. It uses preloaded TypeScript data derived from fixed-seed demonstration records.

18.2 Processing stages

These stages are implementation requirements for the pilot and operating product, not a claim that the current public release performs all nine stages.

Stage	Input	Processing	Output/control
1. Intake	PDF, image, CSV, XLSX, API export	Validate file type, malware, hash, and tenant	Immutable source object
2. Extract	Maintenance record, invoice	Planned AI-assisted OCR and document-field extraction	Candidate value + source location + confidence
3. Resolve	Asset, site, and refrigerant names	Master-data matching and unit normalization	Canonical asset/event IDs
4. Align time	Power, weather, operations	UTC→ICT, interval alignment, missingness flags	Quality-scored time series
5. Analyze	Maintenance, power, weather, factors	Rules, weather baseline, evidence gaps, stress test	Finding candidates
6. Human gate	Candidate, source, checklist	Facility-manager and technician review	Confirmed/rejected state
7. Optimize	Opportunity, cost, constraints	Multi-objective budget selection and sensitivity	Action portfolio

Stage	Input	Processing	Output/control
8. Measure	Action and post-action data	Locked counterfactual, uncertainty, exclusions	Provisional M&V
9. Report	State, formula, version, evidence	Generate executive/technical/ESG views	PDF/JSON/audit package

18.3 Core data object

```
{
  "finding_id": "F-CT-002-2025-04",
  "tenant_id": "TENANT",
  "site_id": "SITE-HCM-CS",
  "asset_id": "CT-HCM-CS-02",
  "finding_type": "refrigerant_risk_and_efficiency",
  "claim_status": "potential",
  "direct_tco2e": 32.623,
  "indirect_tco2": 5.603,
  "factor_basis": ["AR6_Circular_13_2024", "VN_grid_2023"],
  "evidence_ids": ["MR-003", "MR-004", "MR-005"],
  "requires_human_confirmation": true,
  "model_version": "demo-1.0.0"
}
```

The design stores `claim_status`, `factor_basis`, `evidence_ids`, and `model_version` with the number itself.

18.4 Model components

1. Planned AI-assisted OCR and document-field extraction: Generate source-linked candidate asset, refrigerant, mass, date, action-record, and evidence fields for human review. Not implemented in the current release; ranking and numerical decisions remain deterministic.
2. Rules engine: Detect repeated charging, missing recovered mass, evidence gaps, and prohibited language. Demonstrated on fixed pre-authored cases.
3. Weather baseline: Explain healthy electricity use with temperature, humidity, throughput, and door-open time. Demonstrated with a fixed scenario equation, not a fitted customer model.
4. Planned change detection: Use residual magnitude and persistence to identify efficiency-degradation candidates. Not validated on field data.
5. Scenario Engine: Rerun bounded temperature sensitivity and operating conditions. Implemented on the fixed demo case.
6. Action Optimizer: Exhaustively evaluate 32 subsets of five action-ready demonstration cases under one selected objective. Deterministic, not an AI recommendation engine.
7. Claim Engine: Enforce permitted language and verification gates. Demonstrated as deterministic claim controls.

In the initial product, the moat is the evidence object, state transition, and fixed evaluation set, not a large proprietary model.

18.5 Data refresh and version control

Data	Recommended refresh	Version rule
Power and operating data	Every 15 minutes to daily	Source hash, receipt time, missingness rate
Maintenance and refrigerant records	On event	Immutable source; corrections create a new version
NASA POWER weather	Daily append	Store coordinates, time zone, and API parameters
CCKP scenarios	On source update	Lock model, scenario, and period
Grid emission factor	Annual review after official release	State inventory year
GWP and leak-rate factors	On regulatory/reporting change	State AR basis and effective period
Analytical model	After validation and approval	Model card, training period, performance, approver

18.6 Current demo versus operating product

Capability	Current demo	Operating-product target
Data	Public data + deterministic facility scenarios	Real tenant-specific customer data
Intake	Local metadata validation only	Content parsing, schema mapping, source preservation, and production connectors
AI/ML/LLM	None	Evaluated, source-linked OCR/extraction candidates with human review
Users	Public/judge interaction	SSO, roles, approval workflow
Analysis	Fixed formulas and scenarios	Site-specific fitting and model registry
Storage	Static demo assets	Encrypted object, relational, and time-series storage
Audit	Source, formula, and file hash	Immutable log of user, model, and result activity
M&V	15-day scenario replay	Pre-agreed field protocol
Reporting	Pre-generated competition dossier	Dynamic facility-specific PDF/JSON packages

19. Privacy, Security, and AI Governance

Facility energy and maintenance data may contain little personal information, but they can reveal operating patterns, asset vulnerabilities, and production volume. COOL:TRACE therefore treats data minimization, tenant isolation, and auditability as default design requirements.

19.1 Data minimization

- Do not collect technician government IDs, bank accounts, or personal contact details unless analytically necessary.
- Allow optional masking of personal contact information and signatures at intake.
- Define separate retention periods for sources, extracted fields, analytical derivatives, and reports.
- Do not use customer data to train shared models by default.

- Track deletion through the source, derivatives, search indexes, and backup lifecycle.

19.2 Target controls for an operating product

Threat	Control objective	Validation method
Cross-customer disclosure	<code>tenant_id</code> isolation for storage/search and row-level policy	Cross-tenant penetration test
Account takeover	SSO/MFA, least-privilege RBAC, session expiration	Permission regression test and access log
Malicious document	Allow-list types and sizes, malware scan, isolated conversion	Malicious-sample test
Data in transit or at rest	TLS, encryption at rest, managed keys and rotation	Configuration audit and rotation drill
Altered figure or evidence	Source hash, append-only audit log, version signature	Reproducibility and integrity test
Prompt injection	Treat documents as untrusted data, never instructions	Injection red-team set
Overstated AI conclusion	Permitted-language rules, state machine, human gate	Automated prohibited-language evaluation
Excessive retention	Tenant-specific retention and deletion policy	Quarterly retention audit
Service interruption	Backup, recovery objectives, read-only reporting fallback	Recovery drill

These are operating-product requirements. The current static demonstration does not claim that all controls are implemented.

19.3 Role-based access

- Portfolio Admin: Configure sites, users, and factor policy.
- Facility Manager: Review assets, work, and documents.
- Technician: Enter site records for assigned assets.
- ESG/Finance Reviewer: View approved results and reports.
- Independent Reviewer: Read-only access to evidence packages.

Advancing a refrigerant-risk conclusion, changing a baseline, or promoting a result to **verified** requires separation of duties and an approval record—not one user’s unilateral action.

19.4 AI principles

The current release uses no AI/ML/LLM model. The principles below govern only the planned AI-assisted OCR and document-field extraction module. Entity linking, ranking, factors, calculations, optimization, claim-state transitions, and M&V remain deterministic.

1. AI may propose source-linked candidate document fields for human review.
2. AI may not independently establish a leak, safety condition, regulatory compliance, or emissions reduction.
3. Every number must link to a source location or calculation.
4. When confidence is low, request evidence instead of inventing an answer.
5. Preserve model, prompt, and factor versions with the result.
6. Before deployment, confirm applicable Vietnamese privacy, cybersecurity, and refrigerant-handling requirements with local legal and safety experts.

20. Evaluation and Validation Plan

20.1 Offline evaluation available now

The 10 fixed cases are regression-test expectations, not training data. Deterministic calculations, ranking, abstention, and claim-language controls can be tested now. Extraction metrics below are future release gates for the planned AI-assisted module; they are not current performance results.

Evaluation area	Unit	Target gate	Action on failure
Refrigerant, asset, and date extraction	Field F1	≥ 0.95	Disable auto-fill; require manual review
Charge and recovery mass	Exact numeric match	≥ 0.98	Withhold quantification
Claim-to-source linkage	Claim coverage	100%	Block report generation
Unsupported leak-confirmed statements	Count	0	Stop model/rule deployment
Critical-case recall	Recall	100%	Revisit prioritization logic
Critical + High in top five	Recall@5	≥ 0.90	Recalibrate ranking weights
False alarms on normal weather load	CT-004 and CT-009	0/2	Treat as weather-normalization failure
Refusal to quantify insufficient evidence	CT-010	Must pass	Lock result
Calculation reproducibility	Same input/version	100% identical	Review versions and floating point
Temperature-stress monotonicity	+0→+3°C	Energy and peak nondecreasing	Block Scenario Engine release

Ten cases cannot establish generalization. The next stage will create a separate locked set of 50–100 de-identified real documents, stratified by sector, document type, refrigerant, language, and scan quality.

20.2 Three evaluation dimensions

A. Extraction accuracy

- Entity precision, recall, and F1
- Exact match and unit-conversion errors for mass, currency, and date
- Entity-resolution accuracy for aliases of the same asset
- Source bounding-box or sentence-span linkage rate

B. Decision quality

- Priority nDCG@5 and Critical recall
- Share of findings accepted or rejected by technicians
- Unnecessary dispatches and missed high-risk items
- Stability and sensitivity of budget-constrained selections

C. Claim integrity

- Cases in which **potential** is promoted incorrectly to **verified**
- Missing public/demo/model provenance
- Mixed GWP or grid-factor basis
- Direct/indirect double counting

20.3 Weather-baseline evaluation

In a real facility, historical healthy-operation data will be split chronologically into train, validation, and test periods. Random shuffling is not used because it can leak seasonal information.

1. Agree with the technician on the healthy operating window.
2. Define candidate variables: temperature, humidity, operating hours, throughput, and door opening.
3. Lock model structure and exclusions in advance.
4. Evaluate holdout error, bias, and residual stability.
5. Flag extrapolation outside the climate and operating range.

The two M&V tiers are project targets, not current performance claims, and both must be reviewer-approved and pre-locked before outcome review. The minimum outcome-eligibility screen is holdout CV(RMSE) $\leq 25\%$ and absolute NMBE $\leq 10\%$, with data completeness $\geq 90\%$. The stricter Gate-B expansion target is holdout CV(RMSE) $\leq 20\%$ and absolute NMBE $\leq 5\%$. Any adjustment requires documented reviewer rationale; the tiers are not one universal standard.

20.4 Field-pilot design

Stage	Duration	Design	Deliverable
Baseline and mapping	2–4 weeks	Connect asset boundary, power, weather, and maintenance documents	Data-quality and baseline plan
Silent evaluation	2 weeks	Score historical cases without sending alerts	False-positive and miss analysis
Assisted operations	4 weeks	Give technicians evidence and checklists	Acceptance, diagnostic time, action completion
Post-action M&V	4–12 weeks	Compare post-action data with locked baseline	Provisional result and uncertainty
Review	1 week	Joint facility, finance, ESG, and technician review	Go/no-go and expansion scope

20.5 Pilot success criteria and stop conditions

Proposed success criteria

- Asset mapping $\geq 95\%$
- Evidence linkage for material claims 100%
- Technician review of Critical findings 100%
- Upload-to-priority time ≤ 30 minutes
- $\geq 60\%$ of prioritized items approved for action or additional inspection by a technician
- Direct refrigerant reductions remain zero verified claims when mass balance is absent

Stop conditions

- Customer data boundary or consent cannot be confirmed.
- Safety-related high-risk cases are missed repeatedly.
- An unsupported confirmed-leak statement occurs.
- The baseline cannot explain healthy operating conditions.
- A technician judges an action recommendation unsafe.

20.6 Experiment backlog

1. No weather normalization vs. weather normalization: false alarms on CT-004 and CT-009
2. Raw screened-potential lens vs. annual electricity-value lens: budget-driven portfolio differences across the five eligible actions

3. Planned AI-assisted field extraction vs. manual extraction, both source-linked and human-reviewed: exact match, unsupported-field rate, review time, and edit rate
4. 15-day vs. 30-day vs. 90-day M&V: variation and uncertainty
5. On-site weather sensor vs. NASA POWER proxy: baseline error
6. Facility-manager view vs. CFO view: decision time and comprehension

21. Deployment and Commercialization Roadmap

21.1 Days 0–30: Challenge-ready product

Product

- Complete company-grade navigation, Portfolio Command, and the deterministic Action Optimizer; expose access to the pre-generated competition dossier. Dynamic Report Studio remains a pilot module.
- Expose the evidence, value, and state for all 10 assets interactively.
- Add Vietnamese/English display and a mobile review flow.

Data and validation

- Confirm checksums and reproducible builds for all public files.
- Automate the 10-case regression set and prohibited-language tests.
- Show Scenario Lab formulas, factors, and sources.

Gates

- Pass build, accessibility, and core-scenario tests.
- Keep CT-010 unquantified end to end.
- Make every material number in the report traceable.

21.2 Days 31–90: Pilot-ready MVP

- CSV, PDF, and image ingestion with a data-quality interface
- Tenant isolation, RBAC, audit logs, and retention policy
- Facility-specific baseline fitting, locking, and model card
- Mobile technician checklist and approval workflow
- Pricing and value interviews aimed at securing the first paid pilot

Entry gate: Security review, data-processing agreement, asset and meter boundaries, and technician responsibility must be documented.

21.3 Months 3–6: Multi-site proof

- Compare portfolio performance across at least three sectors.
- Measure action acceptance, time to diagnosis, and post-action data completeness.
- Operate grid-factor, GWP, and model registries.
- Test a multi-customer Action Inbox for HVAC/R partners.
- Produce customer-specific executive, technical, and ESG report templates.

Field evaluation must include normal load, insufficient evidence, and low-priority cases—not only high-value anomalies.

21.4 Months 6–12: Commercial scale

- API, SFTP, and BMS-export connectors
- Usage metering and site/asset overage billing

- SLAs, backup and recovery, security testing, and support operations
- Service-delivery model with Vietnamese HVAC/R providers
- Read-only Evidence Room for independent reviewers

21.5 Expansion sequence

1. HCMC/Binh Dương cold storage and food processing: Closest fit to the demonstrated data problem
2. Supermarket portfolios: Repeated asset structures and R-404A management opportunity
3. Hotels and commercial AC: Weather and occupancy normalization across multiple sites
4. Pharmaceutical and health-care cooling: Later-stage market requiring greater reliability and regulatory readiness

This is an execution hypothesis based on data complexity and product readiness—not a market-share claim.

21.6 Capability plan

Capability	Initial responsibility	Addition at scale
Product and customer development	GEONOS product lead	Sector-specific product managers
Data and climate analysis	Data/climate engineer	MLOps and time-series engineers
HVAC/R judgment	External licensed-technician review	Local service-partner network
MRV and carbon	Methodology reviewer	Independent verification partner
Security and privacy	Design owner	External penetration testing and compliance

GEONOS Infrastructure does not currently claim a specific partner, technician review, or pilot commitment.

22. Winning Judge Narrative and Pitch Script

22.1 The sentence judges should remember

Cooling facilities do not lack data. Their weather, power, maintenance, and refrigerant records simply cannot prove anything together. COOL:TRACE turns those records into action and verification.

22.2 Why now

- HCMC public weather data report T2M_MAX of 41.07°C on April 30, 2025.
- Vietnam’s national median CDD65 scenario under SSP2-4.5 for 2040–2059 is approximately 17.8% above the historical period.
- Servicing represented a reported 69% of Vietnam’s 2022 HFC consumption in CO₂e terms.
- Cooling creates both indirect electricity emissions and direct high-GWP refrigerant emissions.
- Many facilities can create value by connecting existing maintenance and power records before adding sensors.

Each statement is national/sector context or a derivative of public data. None is a specific customer outcome.

22.3 Competitive differentiation

Alternative	What it does well	Remaining gap	COOL:TRACE role
BMS/IoT monitoring	Real-time sensors and alarms	Refrigerant documents, source evidence, carbon state	Connect existing data and maintenance evidence
Predictive-maintenance AI	Anomaly and failure probability	Weather false alarms, verification state, direct emissions	Human-gated evidence workflow
Carbon-accounting platform	Organization-wide emissions inventory	Which asset to repair	Asset-level action and M&V
Energy audit	Deep expert diagnosis	Continuous monitoring and scale	Continuous workflow before and after an audit
HVAC maintenance management	Work orders and service history	Climate/power baseline and portfolio optimization	Amplify technician work with data

COOL:TRACE is not a monolithic replacement for these systems. It is the evidence-to-action layer between them.

22.4 Defensibility

The early moat is not model size.

1. A standard data contract linking refrigerant, electricity, weather, and maintenance
2. A workflow that enforces claim states and evidence gates
3. A field-oriented evaluation set that includes normal load and insufficient evidence
4. An M&V ledger that accumulates actual actions and post-action outcomes
5. A network workflow in which facility, technician, finance, and ESG teams use different views of the same evidence

As the customer base grows, the defensible learning asset is not “more leak photos.” It is the record of which evidence led to which action and verification result.

22.5 60-second elevator pitch

Vietnam’s cooling systems use more electricity as temperatures rise, while mishandling a high-GWP refrigerant can create a large climate impact. Today, power data sit in a BMS, refrigerant quantities sit in maintenance records, and heat sits in weather data. No proprietary hardware is required for COOL:TRACE to start; targeted metering may be required when evidence cannot support a decision or M&V. The current interactive prototype demonstrates the intended decision flow on preloaded, deterministic records: it filters load explained by weather, surfaces repeated charging and evidence gaps, and selects demonstration actions under a constrained budget. It never conflates pre-action potential, technician confirmation, post-action modeled estimates, and field verification. Three assets account for 88.8% of the demo’s screened potential, while one remains unquantified because evidence is missing. The pilot will add evaluated AI-assisted OCR and document-field extraction plus real facility M&V; ranking and numerical decisions remain deterministic. COOL:TRACE is not a carbon dashboard; it is an evidence-to-action operating layer for cooling.

22.6 Five-minute live-demo script

Time	Screen/action	Message
0:00–0:25	GEONOS Infrastructure home; 41.07°C, +0.220°C/decade, 60,656 rows	"Heat is not background context. It is an operating condition for cooling infrastructure."
0:25–0:55	Portfolio Command: 5 sites, 10 assets, 175.717 tCO ₂ e potential	"This total is the size of the opportunity to investigate—not a realized reduction."
0:55–1:25	Change budget from VND 25M to VND 60M	"We are not simply recommending more spending. We show which evidence and action the same budget can buy first."
1:25–1:55	Show CT-008, CT-002, CT-007 as top three	"Three assets concentrate 88.8% of screened potential."
1:55–2:35	Open CT-002 MR-003/004/005; 2.8 + 4.1 kg; GWP 4,728	"This is not an AI-confirmed leak. It is a repeated-charge signal that requires technician confirmation."
2:35–3:05	Open CT-010 evidence gap	"Without a scale photo and cylinder reconciliation, the system will not convert 5 kg into tCO ₂ e. Refusing to create a number is a product capability."
3:05–3:40	Scenario Lab +0→+3°C	"For the same operating window, modeled electricity rises 11.59%, from 18,453.7 to 20,593.0 kWh, and 2.25 hours above the 90%-of-nameplate alert threshold appear."
3:40–4:15	CT-002 M&V: signal vs. locked counterfactual	"Over 15 days, the result is 2,267.5 kWh and 1.495 tCO ₂ modeled avoided. Uncertainty is ±12%, and direct refrigerant avoidance remains unverified."
4:15–4:40	Open the pre-generated competition dossier	"The current demo exposes the evidence, formula, and claim state; dynamic facility reports are a pilot module."
4:40–5:00	Company and roadmap	"Trace the evidence. Prioritize cooling action. Prove what changed."

22.7 Demo fallback

- Every core scene has a fixed-data fallback that works without public network access.
- Slider outputs and formulas reproduce from the same versioned JSON.
- The current demo does not upload or parse content. If navigation fails, reset to the preloaded CT-002 and CT-010 cases.
- Prepare the report PDF and a 90-second backup video or screen capture.
- Keep the **synthetic demo** label visible; emphasize the realism of the public data, calculations, and verification design.

22.8 Expected questions and answers

Q. Why should we trust the product without customer data? A. The current evidence supports the data contract, traceability, reproducibility, and overclaim controls—not field performance. That is why the 10-case evaluation set includes normal load and insufficient evidence. The next gate is blind evaluation on real records followed by technician review.

Q. Can NASA POWER diagnose facility failure? A. No. NASA POWER supplies a weather covariate and climate context. Facility sensors, maintenance records, and technician judgment confirm failures and leaks.

Q. How is this different from predictive maintenance? A. COOL:TRACE connects failure screening to direct and indirect climate impact, source evidence, budget allocation, and post-action verification state in one workflow.

Q. What does the hardware-qualified entry mean? A. No proprietary hardware is required to start. It does not mean sensors are unnecessary; targeted metering may be required when existing evidence cannot support a decision or M&V.

Q. Has COOL:TRACE reduced 175.717 tCO₂e? A. No. That is screened potential across nine quantifiable demo cases. The current modeled post-action result is 2,267.5 kWh over 15 days for CT-002. Avoided direct refrigerant emissions remain unverified.

23. Conclusion

COOL:TRACE is not a system that claims to solve every refrigeration and air-conditioning problem in Vietnam automatically. Its role is narrower and more useful: connect fragmented records, reduce false certainty, allocate a limited maintenance budget to the strongest evidence, and prepare post-action performance for credible verification.

GEONOS Infrastructure's core value proposition is:

Turn climate data into facility context, maintenance records into an action sequence, and repair outcomes into verifiable net-zero evidence.

The current demonstration establishes the workflow, provenance controls, calculation transparency, and claim boundaries. It does not establish field reductions. Preserving that distinction while adding real facility data and technician confirmation is the fastest route to a trustworthy climate-cooling service.

Appendix A. Key Figures

Metric	Value	Classification
HCMC mean-temperature OLS trend, 1991–2025	+0.220°C/decade	Public-data derivative
Bàu Bàng mean-temperature OLS trend, 1991–2025	+0.236°C/decade	Public-data derivative
HCMC T2M_MAX, April 30, 2025	41.07°C	NASA POWER public modeled/reanalysis
CDD65, historical	4,133.28°F·day	CCKP national spatial average
CDD65, SSP2-4.5, 2040–2059	4,868.72°F·day	CCKP national scenario
CDD65, SSP5-8.5, 2080–2099	6,447.53°F·day	CCKP national scenario
Vietnam 2023 grid emission factor	0.6592 tCO ₂ /MWh	Official public factor
R-404A AR6 GWP100	4,728	Circular 13/2024
CT-002 direct-emissions potential	32.623 tCO ₂ e	Demo records × official factor
CT-002 indirect-emissions potential	5.603 tCO ₂	Demo assumption × official factor
CT-002 total screened potential	38.226 tCO ₂ e	Demo prioritization value
CT-002 modeled avoided electricity	2,267.5 kWh	Demo M&V model
CT-002 modeled Scope 2 avoided	1.495 tCO ₂	Demo M&V model
CT-002 avoided direct refrigerant emissions	Not verified	Cannot verify from short electricity window
Portfolio baseline annual electricity	4,473,000 kWh	10 demo assets
Portfolio annual electricity opportunity	94,200 kWh	Demo assumption
Portfolio direct screened potential	113.621 tCO ₂ e	Demo records × official GWP
Portfolio indirect screened potential	62.097 tCO ₂	Demo electricity opportunity × official factor
Portfolio quantified screened potential	175.717 tCO ₂ e	Not verified reductions
Top-three concentration	155.963 tCO ₂ e, 88.8%	CT-008, CT-002, CT-007
Proposed pilot outcome target	Normalized cooling energy ≥10%	Nine-month plan; two anchors first, conditional expansion to six; may not be achieved

Appendix B. Primary Data Files

- [vietnam_data/README.md](#) — Bundle scope and interpretation limits
- [PROJECT_DATA_BRIEF.md](#) — Calculation principles and high-integrity demo narrative
- [source_manifest.json](#) — Sources, licenses, and spatial/temporal scope
- [climate_projection_vietnam_cmip6.csv](#) — National climate scenarios
- [climate_stress_test.csv](#) — Temperature-sensitivity experiment
- [maintenance_records.csv](#) — Demo maintenance evidence
- [post_repair_verification.json](#) — Demo post-action M&V
- [calculation_assumptions.json](#) — Formulas and claim boundaries
- [DATA_DICTIONARY.md](#) — Column-level provenance

Appendix C. One-Page Executive Impact Factsheet

GEONOS Infrastructure · COOL:TRACE Vietnam

Product A B2B decision service for cooling records, public climate data, and official factors. The current release uses preloaded deterministic records; parsing, persistence, new-facility analysis, and dynamic reporting are planned pilot modules.

Why now

- HCMC public-data T2M_MAX on April 30, 2025: 41.07°C — PUBLIC
- Vietnam CDD65 under SSP2-4.5 in 2040–2059: +17.8% versus historical — PUBLIC national scenario
- Track 1 expressly includes energy efficiency, carbon measurement, and cooling systems — [Official Challenge Statement](#)

Operating promise [Trace the evidence](#) → [Prioritize cooling action](#) → [Prove what changed](#).

Prototype evidence—not field impact

Metric	Value	Claim state
Public weather rows	60,656	PUBLIC
Portfolio	5 sites · 10 assets · 23 records	DEMO
Annual electricity opportunity	94.2 MWh	DEMO assumption
Direct + indirect screened potential	175.717 tCO ₂ e, calculated from unrounded direct and indirect components	MODEL potential
CT-002, 15-day post-action window	2,267.5 kWh · 1.495 tCO ₂ · ±12%	modeled avoided
Avoided direct refrigerant emissions	Not verified	not verified

Why it is different

1. Adaptation + mitigation: Join heat-driven peak demand and headroom with refrigerant and electricity emissions.
2. Evidence first: Trace every number to its source, formula, factor, and claim state. Planned AI is limited to OCR and field extraction; ranking and numerical decisions remain deterministic.
3. Refusal + constrained economics: Keep CT-010 unquantified when evidence is missing, and show budget-sensitive action portfolios.

Nine-month Vietnam pilot target—planned, not secured

- Two anchor sites first; expand to six only after data, safety, technician-review, M&V, partner, and quotation gates pass
- Target ≥10% reduction in weather- and operations-normalized cooling electricity at eligible, action-completed facilities
- M&V minimum eligibility: holdout CV(RMSE) ≤25% and absolute NMBE ≤10%; stricter Gate-B target: ≤20% and ≤5%; both reviewer-approved and pre-locked
- Evidence linkage 100%; Critical review 100%; unsafe recommendations 0
- Paid continuation target: at least two portfolio contracts covering at least three pilot facilities
- Current local partners, pilots, revenue, and verified field reductions: zero

Commercial hypothesis—not validated

Up to five sites and 100 assets: VND 25M setup + VND 10M/month; revalidate price, delivery cost, and willingness to pay through the pilot.

Challenge fit and ask Track 1 · Prototype/MVP in the program’s TRL 4–5 category; evidence closest to TRL 4. Secure facility data, qualified HVAC/R review, and independent M&V to convert [potential](#) into [verified field reduction](#) through evidence, not assertion.

[Official site](#)